④ Obtenha a transformada de Laplace das funções

a) $f(t) = t e^{-3t} \cos 3t$

$$\mathcal{L}_0[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}\left(\mathcal{L}_0[f(t)]\right)$$

$$= -1 \frac{d}{ds}\left(\mathcal{L}_0[e^{-3t}\cos 3t]\right)$$

(*)

(*) $\mathcal{L}_0[e^{-3t}\cos 3t] = \mathcal{L}_0[e^{at} f(t)] = F(s-a)$

$$\mathcal{L}_0[\cos 3t](s+3)$$

$$\left(\frac{s}{s^2+9}\right)(s+3) = \frac{(s+3)}{(s+3)^2+9}$$

Continuando: $f = s+3$ $\quad g = (s+3)^2 + 9$

$$-1 \frac{d}{ds}\left[\frac{s+3}{(s+3)^2+9}\right] = \left(\frac{f}{g}\right)' = \frac{f' \cdot g - g' \cdot f}{g^2}$$

$(s+3)' = 1$

$((s+3)^2+9)' = 2(s+3)$

$$\frac{1.((s+3)^2+9) - 2(s+3).(s+3)}{((s+3)^2+9)^2} = \frac{s^2+6s+18-2s^2-12s-18}{(s^2+6s+18)^2}$$

$$= \boxed{\frac{-s^2-6s}{(s^2+6s+18)^2}}$$

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b) $f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 2-t, & t \geq 1 \end{cases}$

$g(t) = t$

$h(t) = 2-t$

Solução: Multiplicamos a função $g(t)$ definida em $[0, 1)$, por $u(t) - u(t-1)$ e a função $h(t)$ definida em $[1, \infty)$ por $u(t-1)$. Assim:

$$f(t) = g(t)(u(t) - u(t-1)) + h(t)u(t-1)$$
$$f(t) = t(u(t) - u(t-1)) + (2-t)u(t-1)$$

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c) $f(t) = \sin t \cos(2t)$

Temos que $\cos(\beta)\sin(\alpha) = \frac{1}{2}\left(\sin(\alpha+\beta) - \sin(\alpha-\beta)\right)$

$$= \mathcal{L}_0\left[\frac{1}{2}\left(\sin(2t+t) - \sin(2t-t)\right)\right]$$

$$= \mathcal{L}_0\left[\frac{1}{2}\sin 3t - \frac{1}{2}\sin t\right]$$

$$= \frac{1}{2}\mathcal{L}_0[\sin 3t] - \frac{1}{2}\mathcal{L}_0[\sin t]$$

$$= \frac{1}{2}\cdot\frac{3}{s^2+3^2} - \frac{1}{2}\cdot\frac{1}{s^2+1^2}$$

$$= \frac{3}{2(s^2+9)} - \frac{1}{2(s^2+1)}$$

d) $f(t) = t^2 \underbrace{e^{8t} g(t)}_{h(t)}$

$f(t) = t^2 h(t) \rightarrow \mathcal{L}_0[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}\left[\mathcal{L}_0[f(t)]\right]$

$f(t) = H(t-a) \rightarrow \mathcal{L}_0[f(t)] = \frac{e^{-as}}{s}$

$H(t-8) \rightarrow \mathcal{L}_0[f(t)] = \frac{e^{-8s}}{s}$

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$$(-1)^2 \frac{d^2}{ds^2}\left[\frac{e^{-8s}}{s}\right] = \left(\frac{f}{g}\right)' = \frac{f' \cdot g - g' \cdot f}{g^2}$$

Derivada primeira de $\left(\frac{e^{-8s}}{s}\right)'$

$(e^{-8s})' = -8e^{-8s}$ $\quad s' = 1$ $\quad \rightarrow \frac{(-8e^{-8s})s - 1 \cdot e^{-8s}}{s^2} = \frac{-8se^{-8s} - e^{-8s}}{s^2}$

Derivando novamente, agora $\left(\frac{-8se^{-8s} - e^{-8s}}{s^2}\right)'$

$(-8se^{-8s} - e^{-8s})' = 64e^{-8s}s$

$(s^2)' = 2s$

$$= \frac{64e^{-8s}ss^2 - 2s(-8e^{-8s}s - e^{-8s})}{(s^2)^2}$$

$$= \frac{2e^{-8s}(32s^2 + 8s + 1)}{s^3}$$

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